Pearson Edexcel International GCSE

Mathematics A

Modular Unit 2

June 2024 Reconstructed Past Paper

STUDENT WORKBOOK

Adaptive working-space edition

27 adaptive question sections 104 marks

Selected from Linear Higher Papers 1H and 2H

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Elite IGCSE Mathematics • eliteigcse.com

L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

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About this reconstructed paper

Each question section keeps the exact source demand beside practical working space. A source grid, graph, table, or diagram is retained once at a usable size.

Ownership rule: Unit 2 owns a demand when its assessed or terminal statement is Unit 2, even if Unit 1 skills are prerequisites. For example, drawing boundary lines supports the Unit 2 region-of-inequalities demand. This book contains the demands owned by Unit 2; the selection contains 104 marks.

Contents

Question 01

4MA1-1H Q1

MODULAR UNIT 2
3 marks

1 Here are the first four terms of an arithmetic sequence.

1 4 7 10

(a) Find an expression, in terms of n, for the nth term of this sequence.

....
(2)

The nth term of a different arithmetic sequence is $5n + 17$

(b) Find the 12th term of this sequence.

....
(1)

WORKING SPACE

Question 02

4MA1-1H Q3

MODULAR UNIT 2
2 marks

3 Find the highest common factor (HCF) of 72 and 108
Show your working clearly.

....

WORKING SPACE

Question 03

4MA1-1H Q4

MODULAR UNIT 2
3 marks

4 Ava records the number of kilometres she drives each month.

In April, Ava drove 943 kilometres.
This is 15% more than the number of kilometres she drove in March.

Work out the number of kilometres Ava drove in March.

.... kilometres

WORKING SPACE

Question 04

4MA1-1H Q5

MODULAR UNIT 2

4 marks

5 In the diagram, *ABCDE* is a regular pentagon.

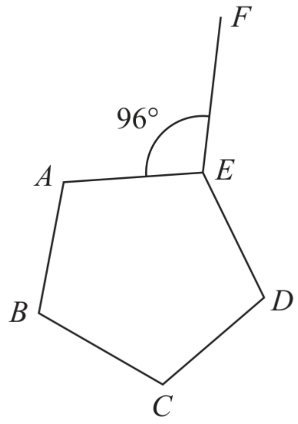


Diagram **NOT** accurately drawn

Angle $AEF = 96°$

Work out the size of the obtuse angle *FED*
Show your working clearly.

.... °

Working space for Question 04

Continue your method

UNIT 2
4 marks

WORKING SPACE

Question 05

4MA1-1H Q9

MODULAR UNIT 2
3 marks

9 The table gives the amount of rice produced by each of two countries in 2020

Country	**Amount of rice (tonnes)**
Indonesia	3.5×10^7
Argentina	8.2×10^5

(a) Write 3.5×10^7 as an ordinary number.

....
(1)

In 2020, Japan produced 6 780 000 more tonnes of rice than Argentina.

(b) Work out the amount of rice Japan produced in 2020
Give your answer in standard form.

.... tonnes
(2)

WORKING SPACE

Question 06

4MA1-1H Q14

MODULAR UNIT 2
5 marks

14 The combined savings of Abel and Bahira are 15 435 dinars.

The savings of Bahira are 45% more than the savings of Abel.

The savings of Bahira are $\frac{3}{2}$ times the savings of Chanda.

Work out the savings of Chanda.

.... dinars

WORKING SPACE

Question 07

4MA1-1H Q15

MODULAR UNIT 2

4 marks

15 The function f is defined as

$$f : x \mapsto \frac{3x + 1}{x - 2}$$

(a) State the value of x that cannot be included in any domain of the function f

....

(1)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ ....

(3)

WORKING SPACE

Question 08

4MA1-1H Q18

MODULAR UNIT 2

5 marks

18 A curve C has equation $y = x^3 - 40x + 1$

Find the coordinates of both the points on C at which the gradient is 8

(.............................. , ...)

(.............................. , ...)

WORKING SPACE

Question 09

4MA1-1H Q21

MODULAR UNIT 2
2 marks

21 A curve has equation $y = f(x)$

There is one minimum point on this curve.
The coordinates of this minimum point are (5, –4)

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 7)$

(.............................. , ...)
(1)

(ii) $y = f(x) - 6$

(.............................. , ...)
(1)

WORKING SPACE

Question 10

4MA1-1H Q23

MODULAR UNIT 2
5 marks

23 A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

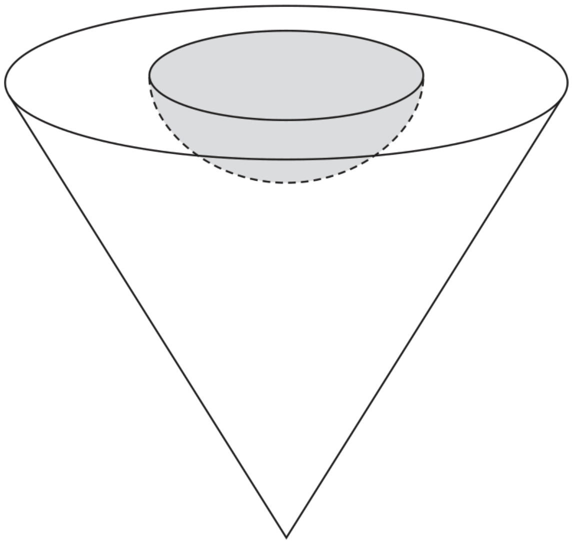


Diagram **NOT** accurately drawn

The radius of the hemisphere is $2x$ cm
The radius of the base of the cone is $5x$ cm
The vertical height of the cone is $6x$ cm

The volume of the solid shape is 6948π cm^3

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.
Give your answer correct to the nearest integer.

.... cm^2

Working space for Question 10

Continue your method

UNIT 2
5 marks

WORKING SPACE

Question 11

4MA1-1H Q24

MODULAR UNIT 2
6 marks

24 A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°
The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.
Show clear algebraic working.

.... °

WORKING SPACE

Question 12

4MA1-2H Q1

MODULAR UNIT 2

3 marks

1 Here are eight numbers written in order of size

$$h \quad 6 \quad 7 \quad 8 \quad j \quad 16 \quad k \quad k$$

where h, j and k are integers.

The median of the eight numbers is 10

The mode of the eight numbers is 18

The range of the eight numbers is 13

Work out the value of h, the value of j and the value of k

h = ...

j = ...

k = ...

WORKING SPACE

Question 13

4MA1-2H Q2

MODULAR UNIT 2
4 marks

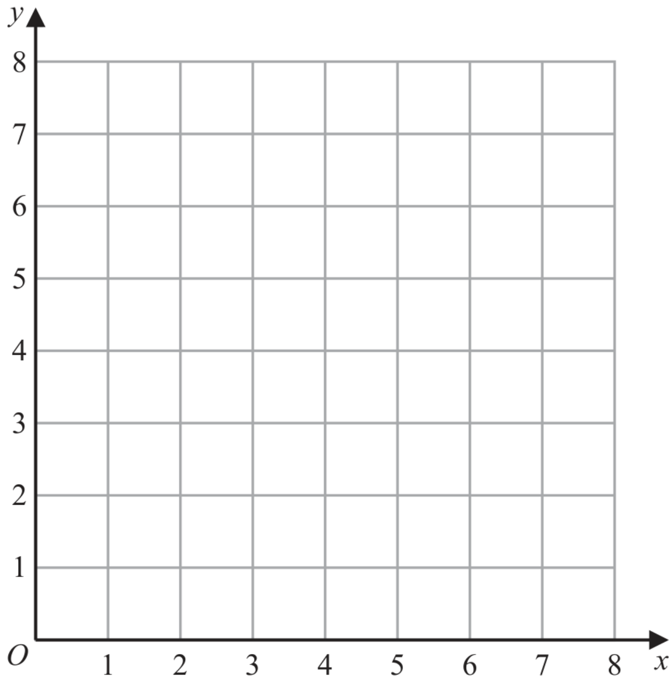
2 (a) On the grid, draw the straight line with equation

(i) $y = 2$

(ii) $x = 6$

(iii) $y = x + 1$

Label each line with its equation.



(3)

(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$$y \geqslant 2 \qquad x \leqslant 6 \qquad y \leqslant x + 1$$

Label the region **R**

(1)

Question 14

4MA1-2H Q8

MODULAR UNIT 2
5 marks

8 Ishir plants 600 bulbs in a garden.
He plants tulip bulbs, crocus bulbs and daffodil bulbs so that

number of tulip bulbs : number of crocus bulbs : number of daffodil bulbs = 9 : 4 : 2

45% of the tulip bulbs are for yellow flowers.

$\frac{5}{8}$ of the crocus bulbs are for yellow flowers.

All of the daffodil bulbs are for yellow flowers.

Work out the number of bulbs that are for yellow flowers.

....

WORKING SPACE

Question 15

4MA1-2H Q9

MODULAR UNIT 2
3 marks

9 Giovanni invests 4500 koruna in a savings account for 4 years.
He gets 2.4% per year compound interest.

Work out how much money Giovanni will have in the savings account at the end of 4 years.
Give your answer correct to the nearest koruna.

.... koruna

WORKING SPACE

Question 16

4MA1-2H Q10

MODULAR UNIT 2

3 marks

10 Solve the simultaneous equations

$$6x + 4y = 1$$
$$3x + 5y = 8$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

WORKING SPACE

Question 17

4MA1-2H Q12

MODULAR UNIT 2
3 marks

12 Ali uses a fitness tracker to count the number of steps he walks each day for 7 days.

For the first 4 days, his mean number of steps is 11 800
For the next 3 days, his mean number of steps is 13 207

Work out his mean number of steps for the 7 days.

....

WORKING SPACE

Question 18

4MA1-2H Q13

MODULAR UNIT 2
7 marks

13 The table gives information about the distances, in km, that 70 teachers travel to school.

Distance (d km)	**Frequency**
$0 < d \leqslant 10$	7
$10 < d \leqslant 20$	17
$20 < d \leqslant 30$	18
$30 < d \leqslant 40$	14
$40 < d \leqslant 50$	10
$50 < d \leqslant 60$	4

(a) Complete the cumulative frequency table.

Distance (d km)	**Cumulative frequency**
$0 < d \leqslant 10$	
$0 < d \leqslant 20$	
$0 < d \leqslant 30$	
$0 < d \leqslant 40$	
$0 < d \leqslant 50$	
$0 < d \leqslant 60$	

(1)

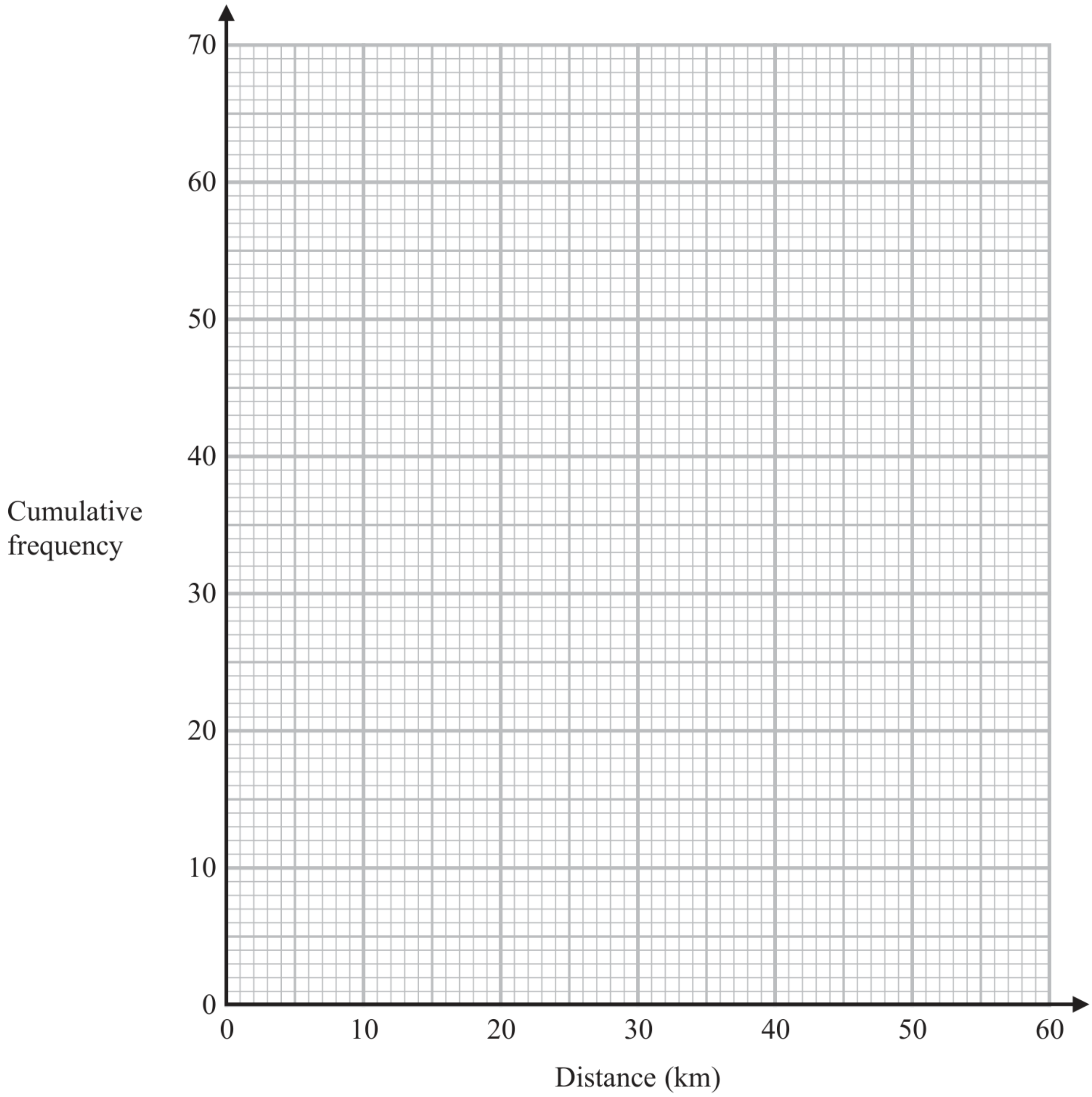
Question 18 (continued)

4MA1-2H Q13

MODULAR UNIT 2
7 marks

(b) On the grid opposite, draw a cumulative frequency graph for your table.

(2)



(c) Use your graph to find an estimate for the interquartile range of the distances.

.... km

(2)

Question 18 (continued)

4MA1-2H Q13

MODULAR UNIT 2
7 marks

(d) Use your graph to find an estimate for the number of teachers who travel more than 46 km.

....

(2)

Question 19

4MA1-2H Q14

MODULAR UNIT 2
3 marks

14 (a) Show that $3y(2y + 5)(y + 7)$ can be written in the form $ay^3 + by^2 + cy$ where a, b and c are integers.

(3)

WORKING SPACE

Question 20

4MA1-2H Q15

MODULAR UNIT 2
4 marks

15 (a) Make g the subject of $e = \sqrt{\dfrac{7g+5}{11+2g}}$

....

(4)

WORKING SPACE

Question 21

4MA1-2H Q15

MODULAR UNIT 2
3 marks

(b) Solve the inequality $3y^2 + 4y - 32 > 0$
Show your working clearly.

....

(3)

WORKING SPACE

Question 22

4MA1-2H Q17

MODULAR UNIT 2

3 marks

17 Q is directly proportional to the square root of d

$Q = 4.5$ when $d = 324$

Find a formula for Q in terms of d

....

WORKING SPACE

Question 23

4MA1-2H Q20

MODULAR UNIT 2

3 marks

20 Given that $k = x - y$ and $x = \frac{1}{4y}$

express $\frac{5k}{x+2}$ in the form $\frac{a - by^2}{c + dy}$ where a, b, c and d are integers.

....

WORKING SPACE

Question 24

4MA1-2H Q21

MODULAR UNIT 2
5 marks

21 The diagram shows a square *ABCD* and a circle.

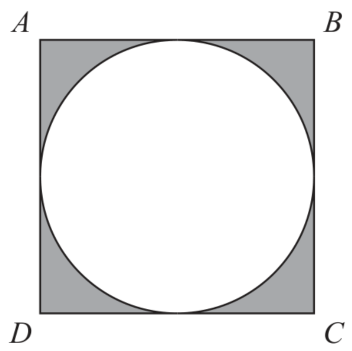


Diagram **NOT** accurately drawn

The sides of the square are tangents to the circle.

The total area of the shaded regions is $80\,\text{cm}^2$

Work out the length of *AC*
Give your answer correct to 3 significant figures.

.... cm

WORKING SPACE

Question 25

4MA1-2H Q22

MODULAR UNIT 2
5 marks

22 The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(.......................... , ...) (.......................... , ...)

WORKING SPACE

Question 26

4MA1-2H Q24

MODULAR UNIT 2
6 marks

24

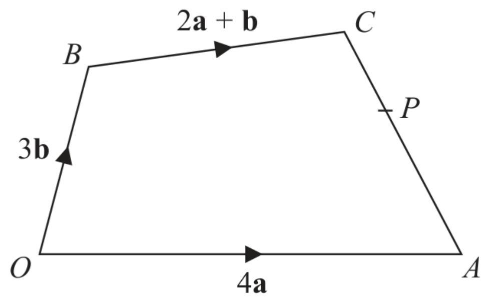


Diagram **NOT** accurately drawn

The diagram shows a quadrilateral *OACB* in which

$\overrightarrow{OA} = 4\mathbf{a}$ $\quad$ $\overrightarrow{OB} = 3\mathbf{b}$ $\quad$ $\overrightarrow{BC} = 2\mathbf{a} + \mathbf{b}$

(a) Find $\overrightarrow{AC}$ in terms of **a** and **b**
Give your answer in its simplest form.

$\overrightarrow{AC} =$ ....

(2)

WORKING SPACE

Question 26 (continued)

4MA1-2H Q24

MODULAR UNIT 2

6 marks

24

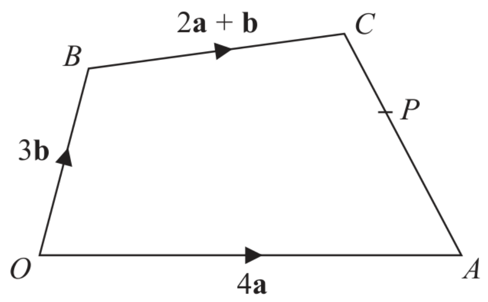


Diagram **NOT** accurately drawn

The diagram shows a quadrilateral *OACB* in which

$\overrightarrow{OA} = 4\mathbf{a}$ $\quad\overrightarrow{OB} = 3\mathbf{b}$ $\quad\overrightarrow{BC} = 2\mathbf{a} + \mathbf{b}$

The point *P* lies on *AC* such that $AP : PC = 3 : 2$

The point *Q* is such that *OPQ* and *BCQ* are straight lines.

(b) Using a vector method, find $\overrightarrow{OQ}$ in terms of **a** and **b**
Give your answer in its simplest form.
Show your working clearly.

$\overrightarrow{OQ} =$ ....

(4)

Working space for Question 26

Continue your method

UNIT 2
6 marks

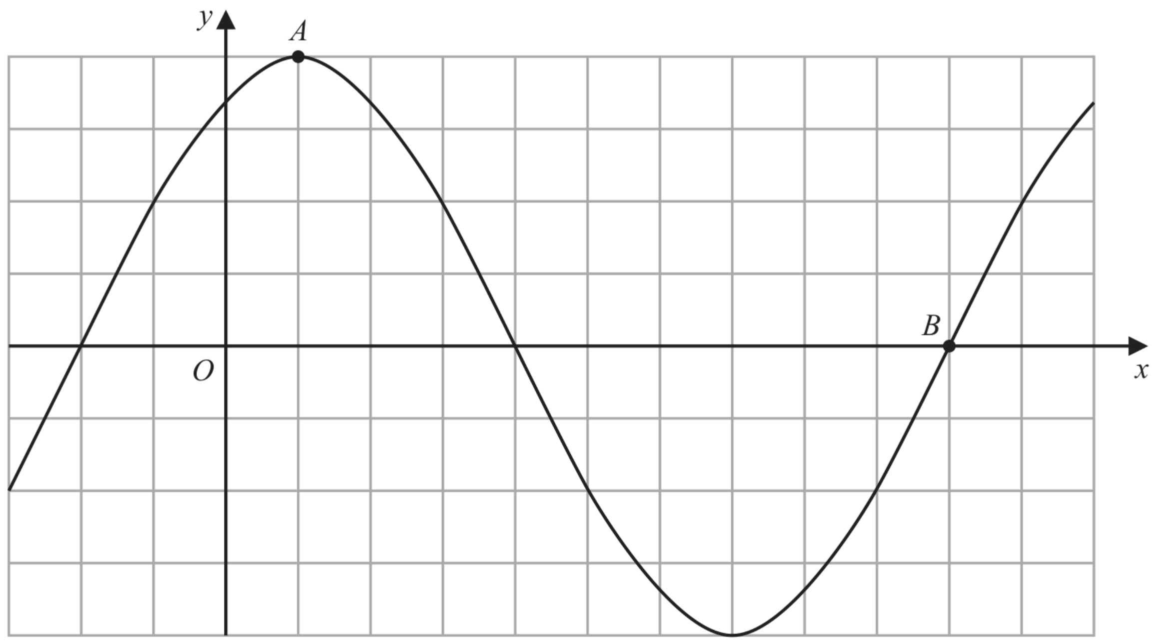
WORKING SPACE

Question 27

4MA1-2H Q25

MODULAR UNIT 2
2 marks

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)°$



(i) Find the coordinates of the point A

(.......................... , ...)

(1)

(ii) Find the coordinates of the point B

(.......................... , ...)

(1)

Working space for Question 27

Continue your method

UNIT 2
2 marks

WORKING SPACE